

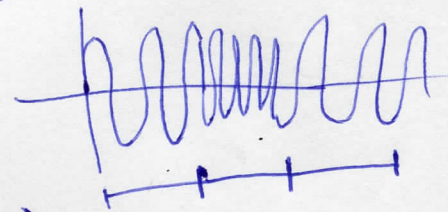
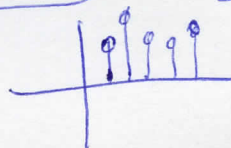
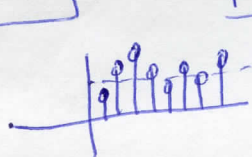
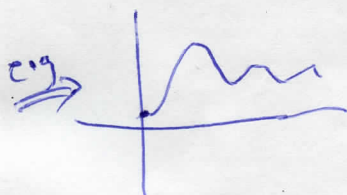
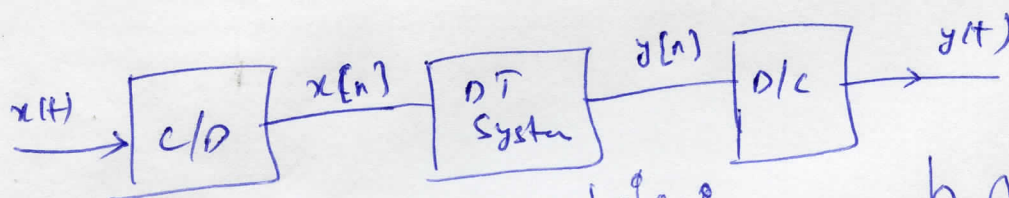
Q. what is a DT system? Input & output DT signals!

- so let's talk about DT signals a while
- repeat some of previous but focusing on DT.

$t \rightarrow n$ in place of t we shall use notation with n

$x[n] \rightarrow n$ discrete e.g. $n = \text{integers}$
 $n = \text{even integer}$ } uniformly spaced
 $n = 0.5K \quad K \text{ integers}$
 $n = 0, 0.1, 0.5, 1, 10, 100$ } non-uniformly spaced

C-D-D-C we often discretize CT signals for easy processing



replace each sample with a sinusoid of different frequency.

Energy & Power

$$x(t) \quad E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$x[n] \quad E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

if $x[n]$ periodic i.e. $x[n] = x[n+N_0] \Rightarrow P_x = \frac{1}{N_0} \sum_{\text{one period}} |x[n]|^2$

Shifting & reversal

$$x_d[n] = x[n-M]$$

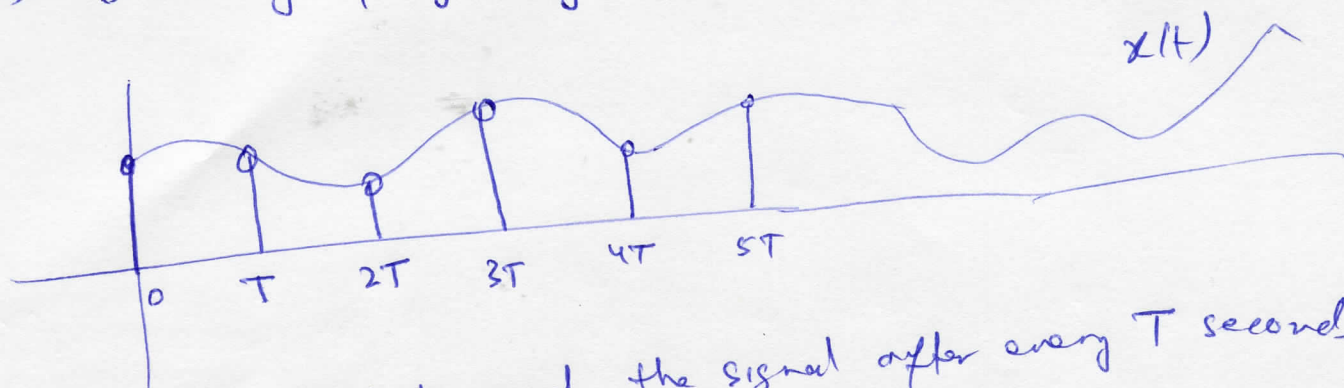
M integer
" "

$$x_a[n] = x[n+M]$$

$$x_r[n] = x[-n] \rightarrow \text{horizontal flip}$$

$$x_v[n] = -x[n] \rightarrow \text{vertical flip}$$

Sampling → our way of getting DT from CT!



→ let's say I want to read the signal after every T seconds.
 → my readings look like this

$$\left(x(t=0) \quad x(t=T) \quad x(t=2T) \quad x(t=3T) \quad x(t=4T) \dots x(t=nT) \right)$$

→ for simplicity I will replace with DT notation + Addition in for of sampling rate (T)

DT version of $x(t)$ → $\left[x[0] \quad x[1] \quad x[2] \quad x[3] \quad x[4] \dots x[n] \dots \right]$

& T given separately eg $T = 0.5 \text{ sec.}$ ← Sampling rate.

Simple Uniform Sampling formula → $x[n] = x(t=nT)$

eg $x(t) = 4t + 5$
 Sample at $T = 0.5 \text{ sec.}$
 then sixth sample is
 $x[6] = x(t=6T)$
 $= x(3) = 12 + 5 = 17$

Downsampling (compression) → let's say you don't want to have samples every 0.5 sec but every 1 sec.

$$x[n] \rightarrow \left[x[0] \quad \cancel{x[1]} \quad x[2] \quad \cancel{x[3]} \quad x[4] \dots \right]$$

$$x_d[n] \rightarrow \left[x_d[0] \quad x_d[1] \quad x_d[2] \dots \right]$$

$x_d[n] = x[2n]$ ← Picked every second sample from DT $x[n]$
 ← so you've Downsampled by a factor of 2!

→ In general, you can downsample by a factor M as

$$x_d[n] = x[Mn] \quad M \text{ integer.}$$

→ what if you wanted to have extra samples in between?
step 1 → upsample (expand)

$$x[n] \rightarrow [x[0] \quad x[1] \quad x[2] \quad x[3] \quad x[4] \quad \dots]$$

$$x_u[n] \rightarrow [x_u[0] \quad 0 \quad x_u[2] \quad 0 \quad x_u[4] \quad 0 \quad x_u[6] \quad \dots]$$

or
$$x_u[n] = \begin{cases} x[n/2] & n = 0, \pm 2, \pm 4, \dots \\ 0 & \text{o/w} \end{cases}$$

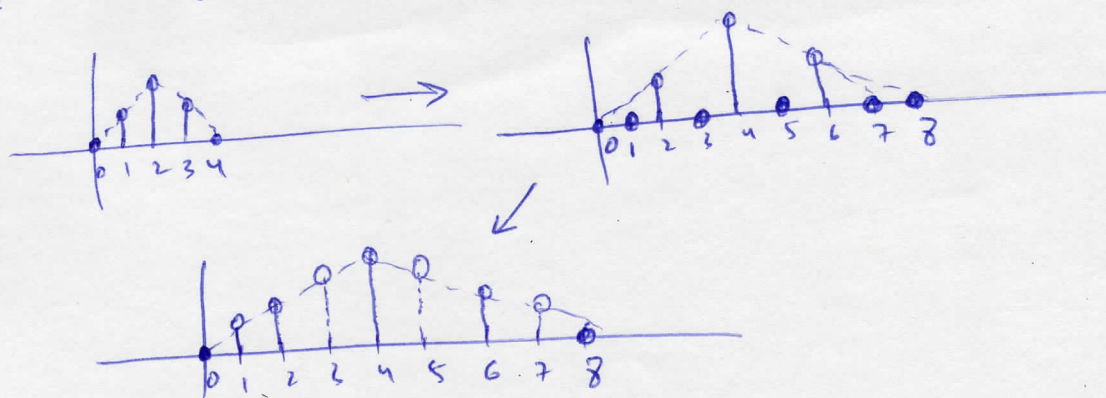
for general upsampling by factor L we have

$$x_u[n] = \begin{cases} x[n/L] & n = 0, \pm L, \pm 2L, \dots \\ 0 & \text{o/w} \end{cases}$$

step 2 replace the inserted zeros by Interpolation.

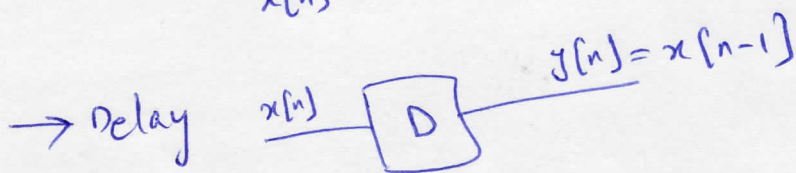
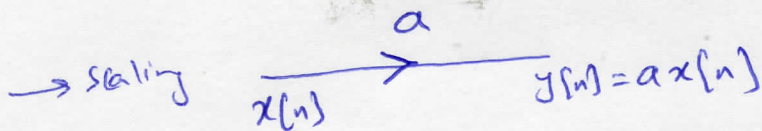
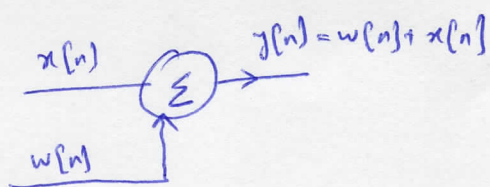
e.g. $x_1[n] \rightarrow [x_u[0] \quad \frac{x_u[0] + x_u[2]}{2} \quad x_u[2] \quad \frac{x_u[4] + x_u[6]}{2} \quad x_u[4] \quad \dots]$

→ Graphically



Denoting DT operations through blocks

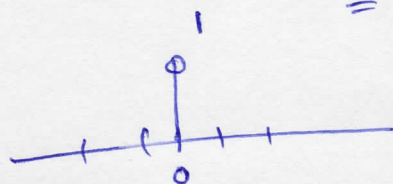
Addition



Some common DT signal models

→ DT Delta

$$\delta[n] = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$$



to draw
Q. Try shifted versions!

→ DT unit step

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

→ DT complex exponential

$$e^{j\Omega n} = \cos(\Omega n) + j \sin(\Omega n) \quad \text{where } \Omega = \text{angular freq}$$

→ link b/w Ω and ω . If a CT sinusoid is sampled every T seconds, we can link the two as

$$\cos(\omega t) \xrightarrow[\text{rate } T]{\text{sample at}} \cos(\Omega n)$$

$$\begin{pmatrix} \cos(\omega t) \\ \rightarrow \cos(\omega n T) \\ \rightarrow \cos(\Omega n) \end{pmatrix}$$

with $\Omega = \omega T$

→ DT exponential

→ we had the CT exponential e^{at}

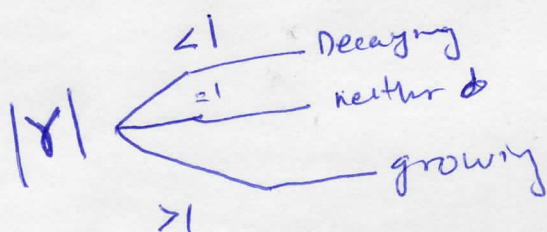
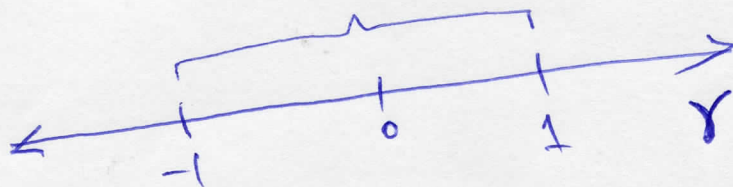
- $a > 0$ growing exponentially
- $a = 0$ constant
- $a < 0$ decaying exponential

→ for DT we have γ^n

Q. Given $n = 0, 1, 2, 3, \dots$

is y^n growing, constant, or decaying?
in magnitude

→ If y is real then



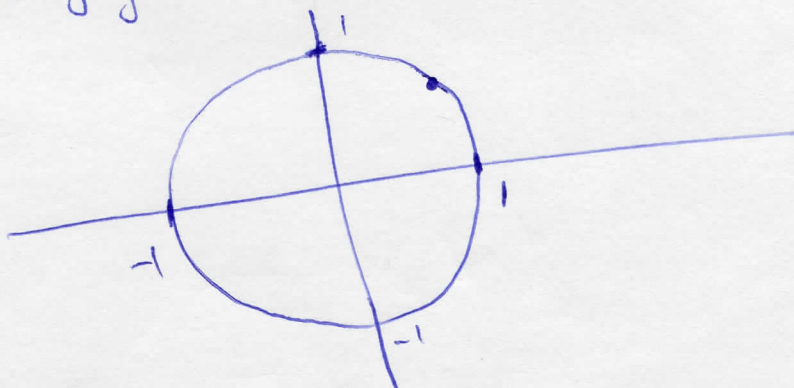
eg
 $(0.5)^n \leftarrow$ ~~growing~~ decay
 $(-0.5)^n \leftarrow$ changing sign but decay
 $(1.5)^n \leftarrow$ grow
 $(1)^n \leftarrow$ neither.

→ Same holds for complex y !

$$y = a + jb \Rightarrow |y| = \sqrt{a^2 + b^2}$$

$|y| < 1$ Decaying etc.

Q. what does this mean in complex plane



Inside unit circle → decaying
 outside unit circle → growing
 on unit circle → neither